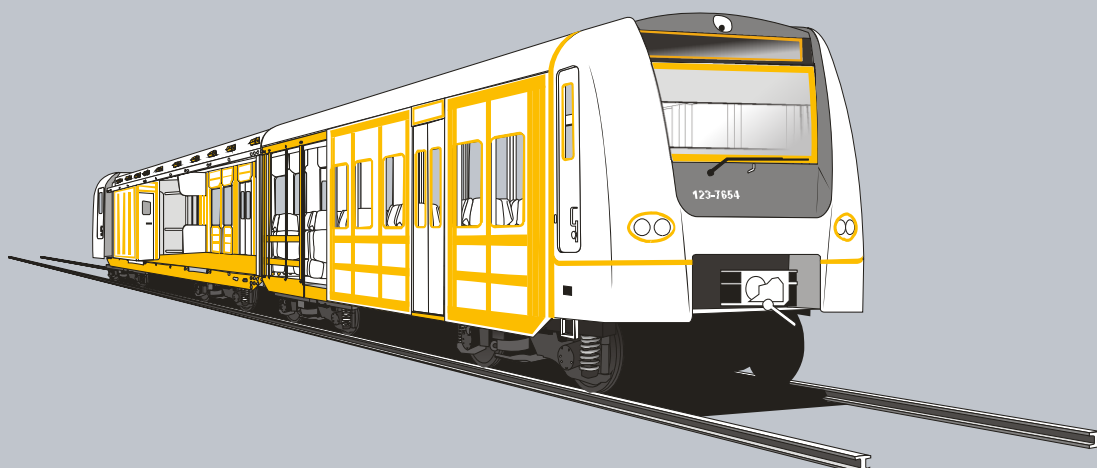




ADDITIONAL PRODUCT INFORMATION

Sikaflex®-265

13.11.2013 / VERSION 3 / SIKA SERVICES AG / MARKETFIELD ENGINEERING

CHEMICAL AND MECHANICAL PROPERTIES ACCORDING E DIN 6701-3: 2010-8

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1 CHEMICAL PROPERTIES

1.1 CHEMICAL BASE

C1	Chemical base	
	Polyurethane	See product data sheet

1.2 CURING MECHANISM

C2	Curing mechanism	
	1C Moisture curing	See product data sheet

1.3 VOLATILE-ORGANIC COMPOUNDS

C3	Volatile Organic Compounds	
	< 0.01% VOC	See safety data sheet / Chapter 15

1.4 TOXICOLOGICAL PROPERTIES

C4	Toxicological properties	
	Classification (67/548/EEC)	See safety data sheet / Chapter 2.1

1.5 PH- VALUE

C5	pH Value	
	According E DIN 6701-3	6.5

Additional Product Information

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2 MECHANICAL PROPERTIES

2.1 DENSITY UNCURED

M1	Density uncured
	See product data sheet

2.2 COLOUR

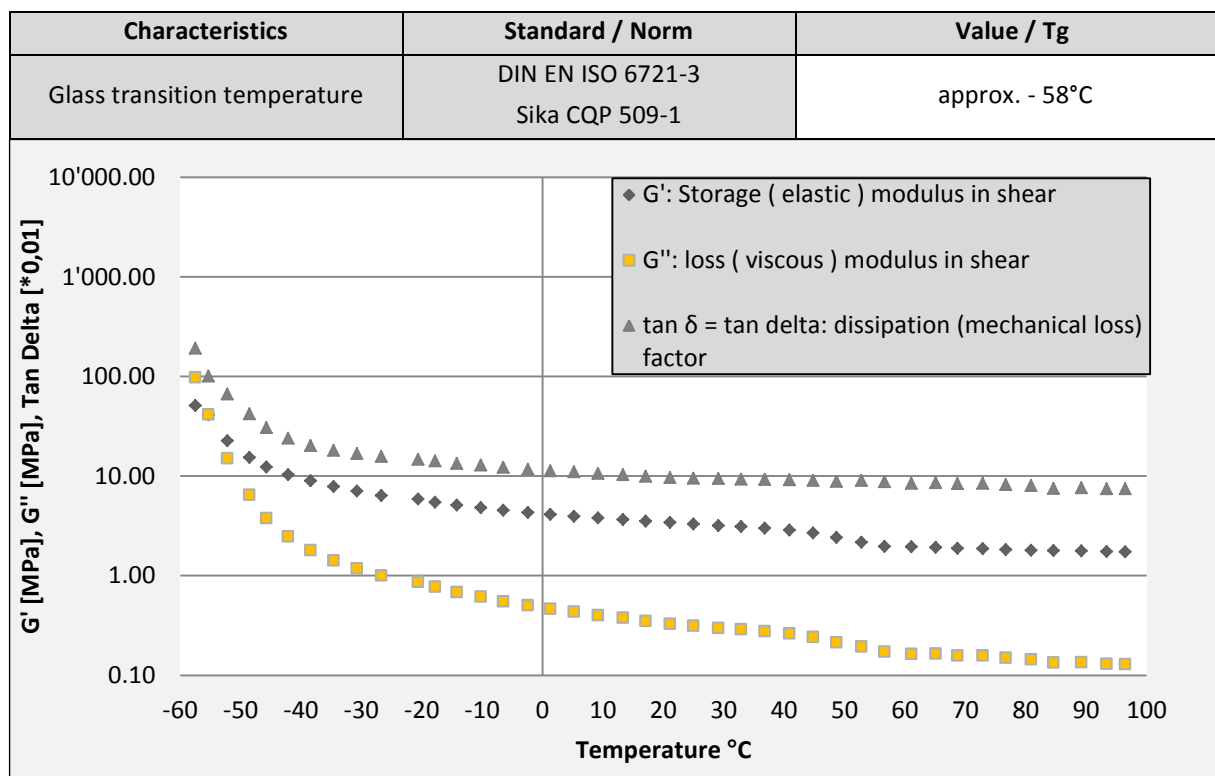
M3	Colour
	black

2.3 SHRINKAGE

M5	Shrinkage
	See product data sheet

2.4 THERMO MECHANICAL BEHAVIOR & GLASS TRANSITION TEMPERATURE

M7	Thermo mechanical behavior & Glass transition temperature	E DIN 6701:2010-8
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Additional Product Information

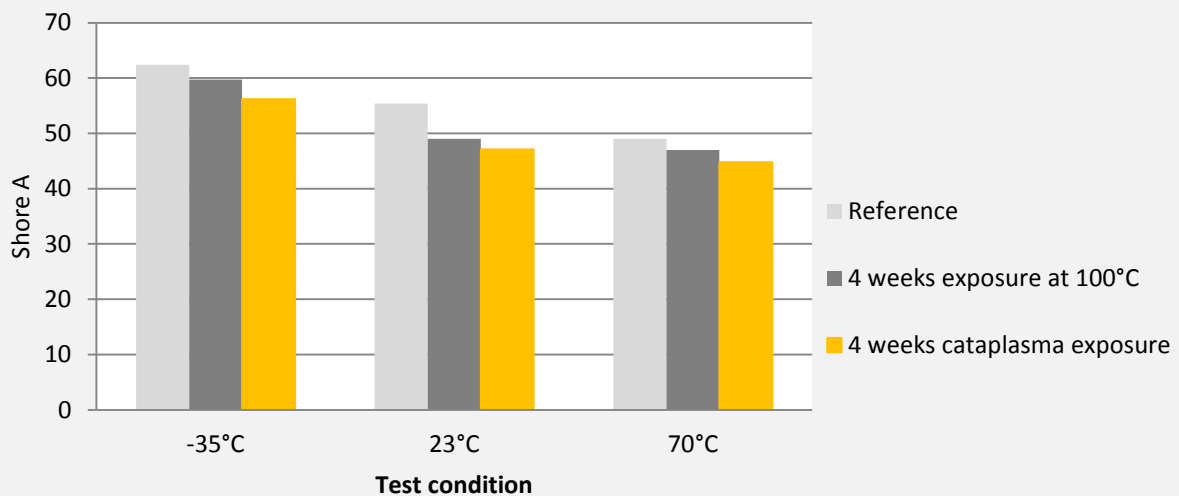
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2.5 SHORE A HARDNESS

M8	Shore A hardness	E DIN 6701:2010-8
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After curing, 7 days at 23°C/50%r.h.			
Characteristics	Standard / Norm	Test condition	Value
Shore A	DIN 53505 Sika CQP 023-1	-35°C	62
		23°C	55
		70°C	49
After 4 weeks cataplasma ageing (70°C/100% r.h.)			
Characteristics	Standard / Norm	Test condition	Value
Shore A	DIN 53505 Sika CQP 023-1	-35°C	56
		23°C	47
		70°C	44
After 4 weeks at 100°C			
Characteristics	Standard / Norm	Test condition	Value
Shore A	DIN 53505 Sika CQP 023-1	-35°C	59
		23°C	49
		70°C	47



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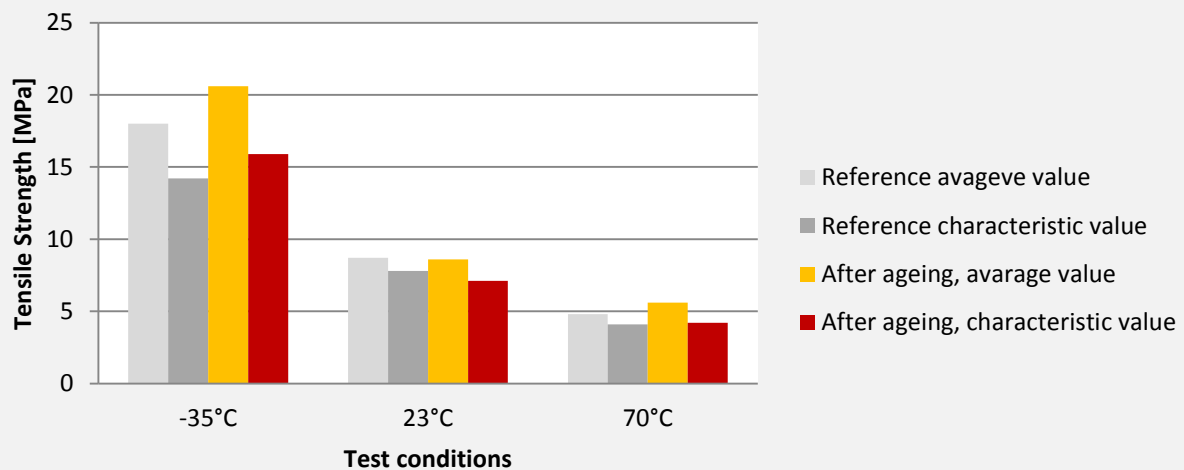
2.6 TENSILE STRENGTH / ELASTICITY / E-MODULUS

M9	Tensile Strength / Elasticity / E-Modulus	Quasi Static Test E DIN 6701:2010-8
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Conditions	Standard average values			Characteristic value R _c	
	Tensile Strength [σ _{max}]	Elasticity [% _{max}]	E-Modulus [E]	Tensile Strength [σ _{max}]	Elasticity [% _{max}]
-35°C	18.3 MPa	404%	10.5 MPa	14.2 MPa	294%
23°C / 50%r.h.	8.7 MPa	488%	3.4 MPa	7.8 MPa	431%
70°C	4.8 MPa	248%	2.6 MPa	4.1 MPa	201%

After ageing cycle according to DIN 54457
7d 23°C/50%r.h. + 7d water immersion + 1d at 80°C + 7d cataplasma (70°C/100%r.h.)

Conditions	Standard average values			Characteristic value R _c	
	Tensile Strength [σ _{max}]	Elasticity [% _{max}]	E-Modulus [E]	Tensile Strength [σ _{max}]	Elasticity [% _{max}]
-35°C	20.6 MPa	377%	18.2 MPa	15.9 MPa	268%
23°C / 50%r.h.	8.6 MPa	403%	2.6 MPa	7.1 MPa	358%
70°C	5.6 MPa	266%	2.2 MPa	4.2 MPa	180%



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2.7 COMPARISON OF MECHANICAL PROPERTIES WITH AND WITHOUT BOOSTER PASTE

		Standard average values		
System	Test condition	Tensile Strength [$\sigma_{\max}$]	Elasticity [% $_{\max}$]	E-Modulus [E]
Reference (without Booster)	- 35°C	19.4 MPa	430%	7.6 MPa
Sika®Booster 20W		14.5 MPa	337%	8.3 MPa
Sika®Booster Paste		18.6 MPa	455%	7.2 MPa
Reference (without Booster)	23°C	9.3 MPa	487%	3.9 MPa
Sika®Booster 20W		9.5 MPa	496%	3.9 MPa
Sika®Booster Paste		8.8 MPa	447%	3.7 MPa
Reference (without Booster)	70°C	5.9 MPa	292%	3.2 MPa
Sika®Booster 20W		5.5 MPa	286%	3.0 MPa
Sika®Booster Paste		5.7 MPa	277%	3.2 MPa

Test condition	without Booster	Sika®Booster 20W	Sika®Booster Paste
-35°C	19.4	14.5	18.6
23°C	9.3	9.5	8.8
70°C	5.9	5.5	5.7

Based on the results listed above we can confirm that the influence of the booster paste on mechanical properties is negligible.

Additional Product Information

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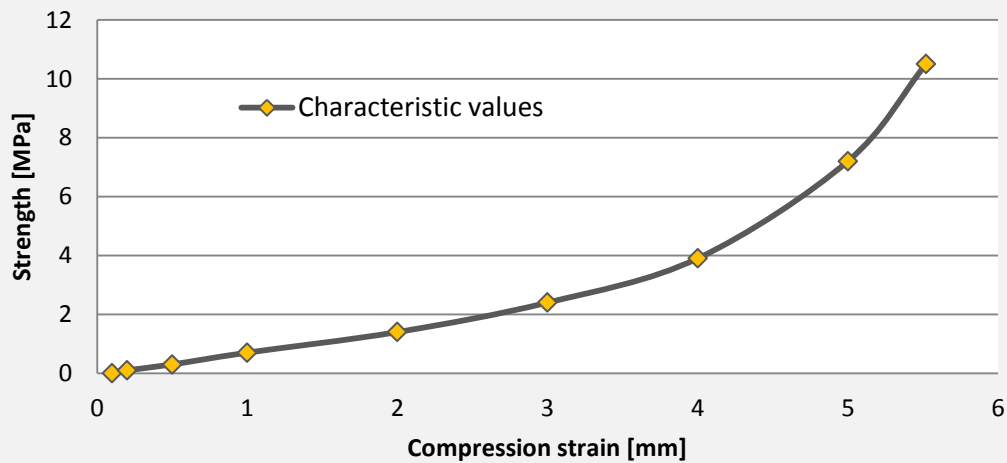
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2.8 COMPRESSION STRENGTH

M10	Compression Strength	Quasi Static Test E DIN 6701:2010-8
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Test Condition	Deformation [mm]	Compression stress [σ]	Compression force [N]
23°C	1	0.7 Mpa	201 N
	2	1.4 Mpa	408 N
	3	2.4 Mpa	682 N
	4	3.9 Mpa	1135 N
	5	7.2 Mpa	2108 N
	5.5	10.5 Mpa	3147 N

Sample dimension (cylinder): 10mm height, 20mm diameter



Additional Product Information

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2.9 LAP SHEAR STRENGTH

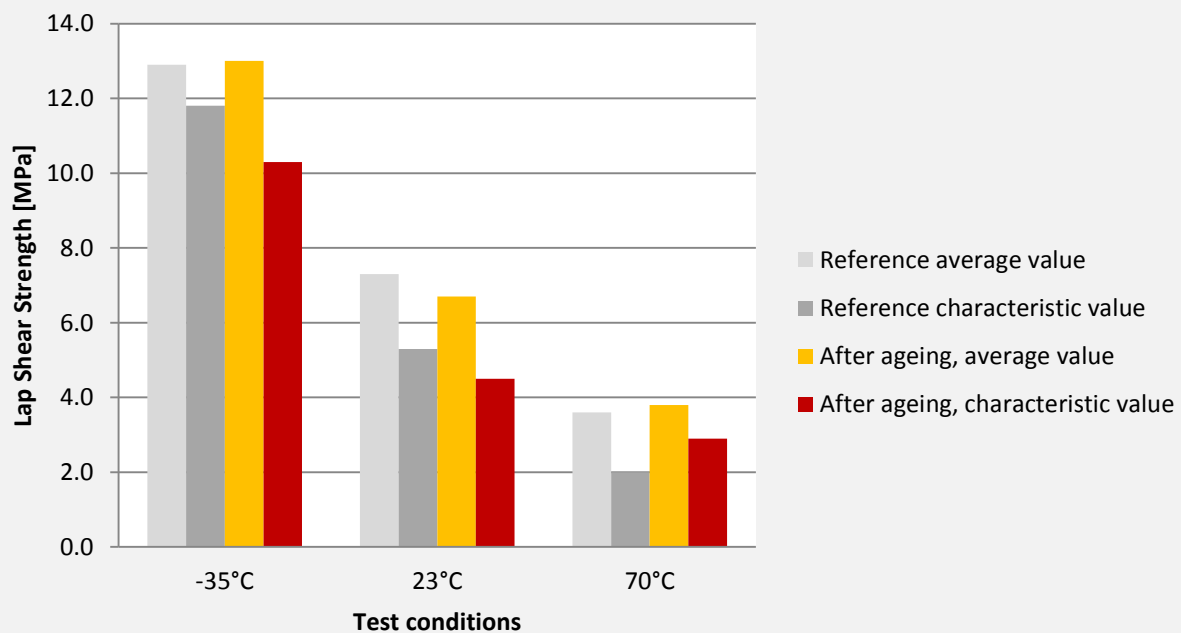
M11	Lap Shear Strength	Quasi Static Test E DIN 6701:2010-8
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	Standard average values		Characteristic value R _c	
Conditions	Lap shear strength [τ _{max}]	Shear strain at break [γ _{max}]	Lap shear strength [τ _{max}]	Shear strain at break [γ _{max}]
-35°C	12.9 Mpa	425%	11.8 Mpa	256%
23°C / 50%r.h.	7.2 Mpa	496%	5.3 Mpa	342%
70°C	3.6 Mpa	338%	2.0 Mpa	142%

After ageing cycle according to DIN 54457
7d 23°C/50%r.h. + 7d water immersion + 1d at 80°C + 7d cataplasma (70°C/100%r.h.)

	Standard average values		Characteristic value R _c	
Conditions	Lap shear strength [τ _{max}]	Shear strain at break [γ _{max}]	Lap shear strength [τ _{max}]	Shear strain at break [γ _{max}]
-35°C	13.0 Mpa	431%	10.3 Mpa	239%
23°C / 50%r.h.	6.7 Mpa	425%	4.5 Mpa	164%
70°C	3.8 Mpa	258%	2.9 Mpa	143%

Dimension of bond line: overlap = 12mm, width = 25mm, thickness = 3mm



Additional Product Information

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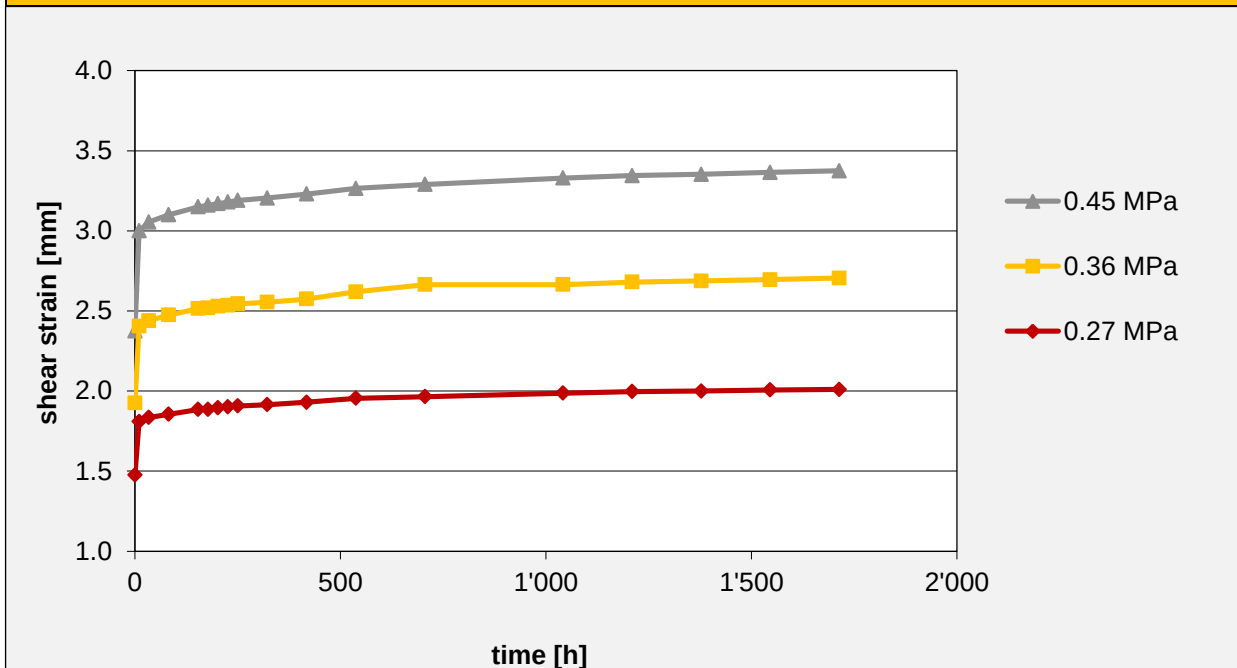
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2.10 CREEP

M12	Creep	Static Test E DIN 6701:2010-8
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Test condition	Static load	Deformation [mm]	Deformation [%]
90 days at 23°C	6.75 Kg / 0.27 MPa	2.0 mm	40%
	13.5 Kg / 0.36 MPa	2.7 mm	54%
	22.5 Kg / 0.45 MPa	3.4 mm	68%

Dimension of bond line: overlap = 20mm, width = 25mm, thickness = 5mm



* 0.27 MPa $\cong$ 6% of lap shear strength value stated in the product data sheet of Sikaflex®-265

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2.11 RELAXATION SHEAR FAILURE STRAIN

M13	Relaxation Shear Failure Strain	Static Test E DIN 6701:2010-8
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Characteristics	Test condition	Deformation	Assessment
Allowable shear strain	1000 hours at 23°C	(tan γ) 0.5 50% = 2.5mm	Visual assessment OK NO FAILURE

After ageing cycle according to DIN 54457 7d 23°C/50%r.h. + 7d water immersion + 1d at 80°C + 7d cataplasma (70°C/100%r.h.)			
Characteristics	Test condition	Deformation	Assessment
Allowable shear strain	1000 hours at 23°C + ageing cycle	(tan γ) 0.5 50% = 2.5mm	Visual assessment OK NO FAILURE
Dimension of bond line: overlap = 20mm, width = 25mm, thickness = 5mm			

Additional Product Information

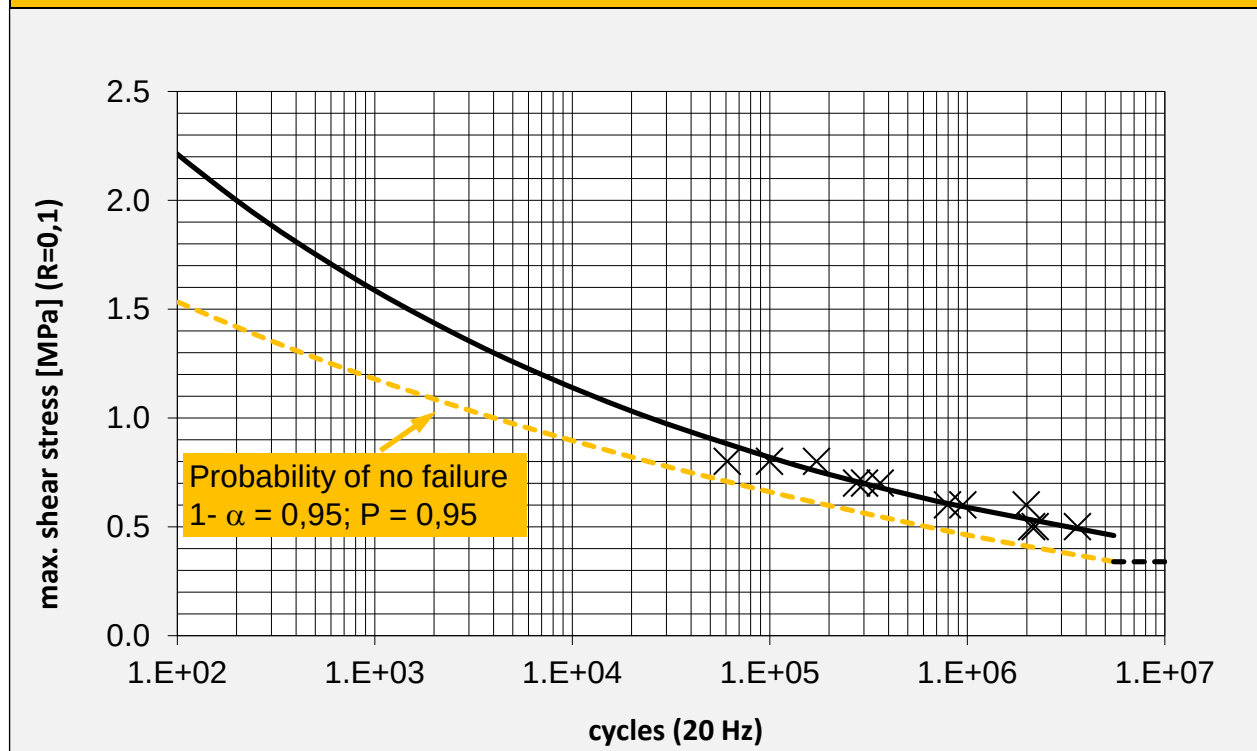
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2.12 SHEAR FATIGUE TESTING

M14	Shear Fatigue Testing	Dynamic Test
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Characteristics	Test condition	Cycles	Nominal shear stress amplitude at cycles to failure
95% Statistical probability of no failure	23°C	10^7	0.34 Mpa
Frequency: According to E DIN 6701:2010-08 Appendix A / Section 1.3.7 Amplitude ratio value / $R = 0.1$ stress regulated amplitude			



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3 REFERENCES

Pos.	Report no: / Source	Title
C1	Product data sheet	Chemical base
C2	Product data sheet	Curing mechanism
C3	Safety data sheet	VOC content
C4	Safety data sheet	Toxicological properties
C5	00506-CTS1-00005-SKh	pH Values of Sikaflex® adhesives according to E DIN 6701-3
M1	Product data sheet	Density uncured
M3	Product data sheet	Colour
M5	Product data sheet	Shrinkage
M7	00509-CTS1-00017-MKö	Thermo mechanical behaviour & glass transition temperature
M8	00509-CTS1-00004-MKö	Shore A hardness of Sikaflex®-265 according to E DIN 6701-3
M9	00509-CTS1-00011-MKö	Tensile strength of Sikaflex®-265 acc. To E DIN 6701-3
	00509-CTS1-00044-MKö	Influence of Booster Paste on mechanical properties of Sikaflex®-265
M10	00509-CTS1-00043-MKö	Compressive strength Sikaflex®-265 acc. to E DIN 6701-3
M11	00509-CTS1-00023-MKö	Lap shear strength of Sikaflex®-265 acc. to E DIN 6701-3
M12	00509-CTS1-00029-BVs	Creep behavior of Sikaflex®-265 acc. to E DIN 6701-3
M13	00509-CTS1-00036-BVs	Relaxation test of Sikaflex®-265 following E DIN 6701-3
M14	2013-05-01-MM-SKo	Fatigue Test of Sikaflex®-265

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